

Algebra PhD Exam, September 1988

Do seven problems, at least two from each section.

I. Group Theory.

- Let G be the group of all 3×3 non-singular matrices of determinant 1 with coefficients in the field of 3 elements.
 - Find $|G|$.
 - Find a Sylow 3-subgroup of G .
 - Compute $|\text{Syl}_3(G)|$, where $\text{Syl}_3(G)$ is the set of all Sylow 3-subgroups of G .
- Let G be a finite group acting transitively on a set X , $x \in X$, and P be a Sylow p -subgroup of G_x for some prime p . Consider the set $\text{Fix}(P) = \{y \in X \mid y^P = y\}$, the set of fixed points of P .
 - Prove that $N_G(P)$ acts on $\text{Fix}(P)$, i.e., maps $\text{Fix}(P)$ to $\text{Fix}(P)$.
 - Prove that $N_G(P)$ acts transitively on $\text{Fix}(P)$.
- Two permutation representations $G \xrightarrow{\pi_i} \Sigma(X_i)$, (the symmetric group on X_i are equivalent if there exists $\alpha : X_1 \rightarrow X_2$ such that for all $g \in G$, $x \in X_1$, we have $x(g\pi_1)\alpha = (x\alpha)(g\pi_2)$). Prove that the following two permutation representations of S_4 of degree 12 are not equivalent:
 - on the right coset space of $\langle(12)\rangle$.
 - on the right coset space of $\langle(12)(34)\rangle$.
- Let G be a finite group and $A \leq \text{Aut}(G)$ such that $G = [G, A]$. Suppose $N \leq G$ satisfying $[N, A] = 1$. Use the 3-subgroup lemma to prove $N \leq Z(G)$, the center of G . (Recall the 3-subgroups lemma: For three subgroups X, Y, Z of a group M , if $[X, Y, Z] = 1$ and $[Y, Z, X] = 1$, then $[Z, X, Y] = 1$.)

II. Homological algebra.

- Let R be a ring and consider the following commutative diagram of R -modules and R -module homomorphisms such that each row is a short exact sequence.

$$\begin{array}{ccccccccc}
 0 & \longrightarrow & A & \xrightarrow{f} & B & \xrightarrow{g} & C & \longrightarrow & 0 \\
 & & \alpha \downarrow & & \beta \downarrow & & \gamma \downarrow & & \\
 0 & \longrightarrow & A' & \xrightarrow{f'} & B' & \xrightarrow{g'} & C' & \longrightarrow & 0
 \end{array}$$

Prove that if α and γ are isomorphisms then β is also an isomorphism.

- Prove the $\mathbb{Z}$ -module isomorphisms, where $c = (m, n)$ is the greatest common divisor.
 - $\mathbb{Z}/m\mathbb{Z} \otimes \mathbb{Z}/n\mathbb{Z} = \mathbb{Z}/c\mathbb{Z}$.
 - $\text{Hom}(\mathbb{Z}/m\mathbb{Z}, \mathbb{Z}/n\mathbb{Z}) = \mathbb{Z}/c\mathbb{Z}$.

3. A module P over a ring R is said to be projective if, given any diagram of R -module homomorphisms with g surjective

$$\begin{array}{ccc} & & P \\ & & \downarrow f \\ A & \xrightarrow{g} & B \end{array}$$

there exists an R -module homomorphism $h : P \rightarrow A$ such that the following diagram is commutative

$$\begin{array}{ccc} & & P \\ & \swarrow h & \downarrow f \\ A & \xrightarrow{g} & B \end{array}$$

Prove that the following conditions on a ring R with identity are equivalent:

- (a) Every R -module is projective.
- (b) Every short exact sequence of R -modules splits.

III. Commutative algebra.

1. A non-empty subset S of a ring R with identity is called multiplicative if $ab \in S$ whenever $a, b \in S$.
 - (a) If I is a proper ideal of R disjoint from S , then prove that there exists an ideal P maximal with respect to containing I and being disjoint from S . Furthermore, show that P is a prime ideal.
 - (b) If P is a prime ideal of ring R , define the localization of R at P and prove that the localization has a unique maximal ideal.
2. Find a reduced primary decomposition for the following ideals. Also give the prime ideals to which the primary ideals belong.
 - (a) (24) as an ideal in $\mathbb{Z}$.
 - (b) $(2x, x^2)$ as an ideal in $\mathbb{Z}[x]$.
 - (c) What theorems allow us to conclude that every ideal in $\mathbb{Z}[x]$ has a primary decomposition.
3. Let R be a unique factorization domain and K its field of fractions.
 - (a) Show that if $a \in K$ is integral over R , then $a \in R$. (In other words, R is integrally closed.)
 - (b) True or false: If S is a subring of K containing R , then S is also a unique factorization domain. If true, prove your answer. If false give examples of R , K , and S that show it not to be true.
4. Determine, up to isomorphism, all semisimple rings of order $1008 = 2^5 \cdot 3^2 \cdot 7$.